

Gene Expression Analyzer

Upload a gene expression CSV file to perform clustering analysis and pattern discovery.

Gene Expression Clustering and Pattern Discovery

Alzheimer's Disease Gene Expression Analysis

This application performs:

- Data normalization
- K-Means clustering
- PCA visualization
- Heatmap analysis
- Hierarchical clustering
- Biological interpretation
- PDF report generation

Upload CSV File



Drag and drop file here

Limit 200MB per file • CSV

Browse files



GSE304521_raw_counts_all_samples.csv 18.3MB

×

Dataset uploaded successfully!

Dataset Preview

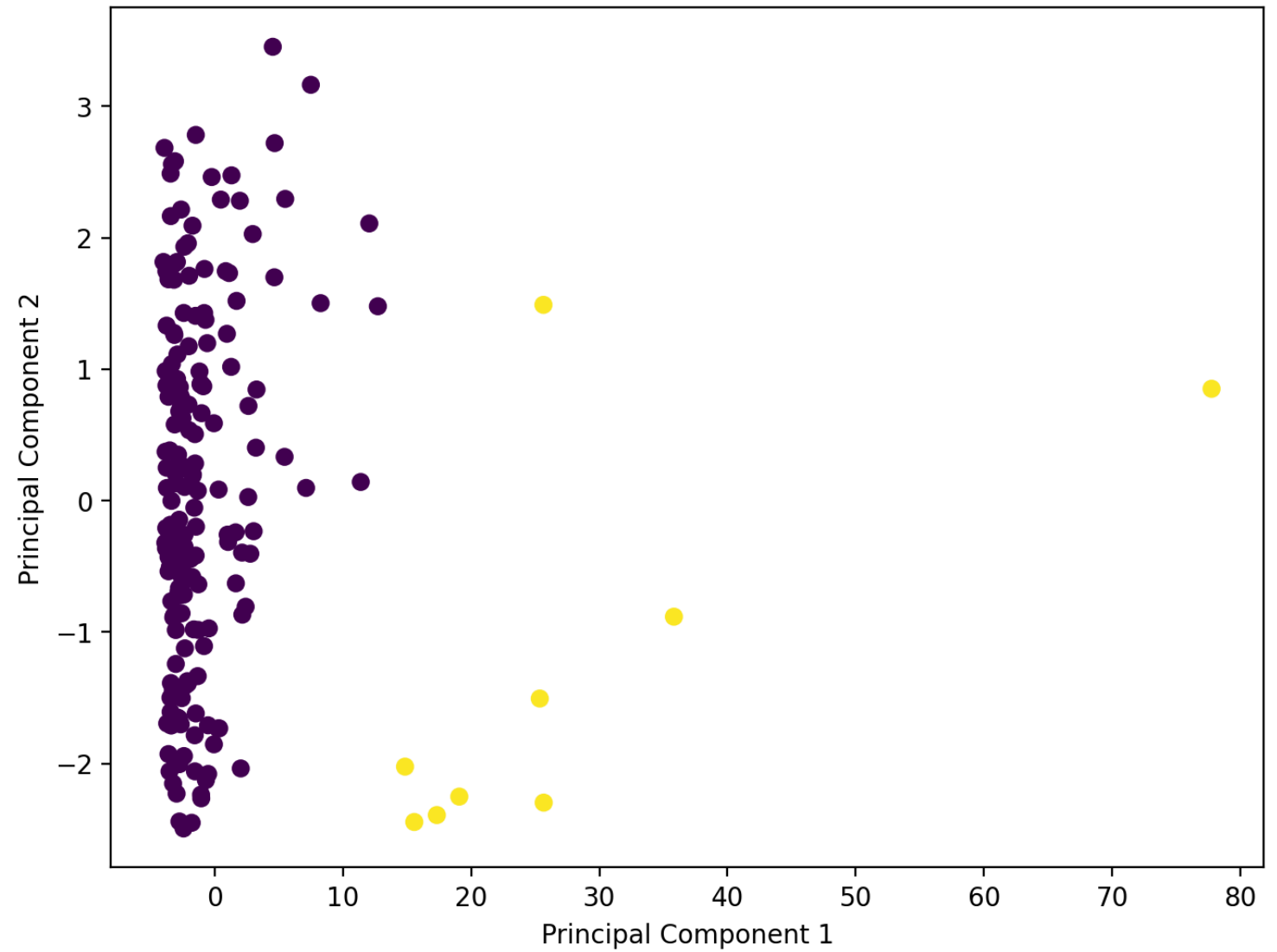
	Unnamed: 0	gene_id	CtrlVeh_1F	CtrlVeh_2F	CtrlVeh_3F	CtrlVeh_4F	CtrlVeh_5M	CtrlVeh_6M	CtrlVeh_7M	CtrlVeh_8M	CtrlC176_1F
0	1	ENSMUSG00000064351	1,753,748	2,228,182	2,026,711	1,888,880	2,171,856	1,652,041	1,543,294	1,623,285	1,600,815
1	2	ENSMUSG00000064370	932,869	1,163,853	1,125,058	1,044,255	1,098,402	900,978	759,169	827,920	815,067
2	3	ENSMUSG00000064339	580,454	785,521	792,675	754,945	799,538	611,839	487,016	518,059	514,056
3	4	ENSMUSG00000064341	695,799	903,452	866,766	781,624	834,631	671,611	570,795	614,972	620,296
4	5	ENSMUSG00000119584	911,748	965,561	1,194,975	519,594	729,044	335,076	439,669	969,073	591,864

Normalization Completed

Cluster Results

	Cluster
0	1
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	0

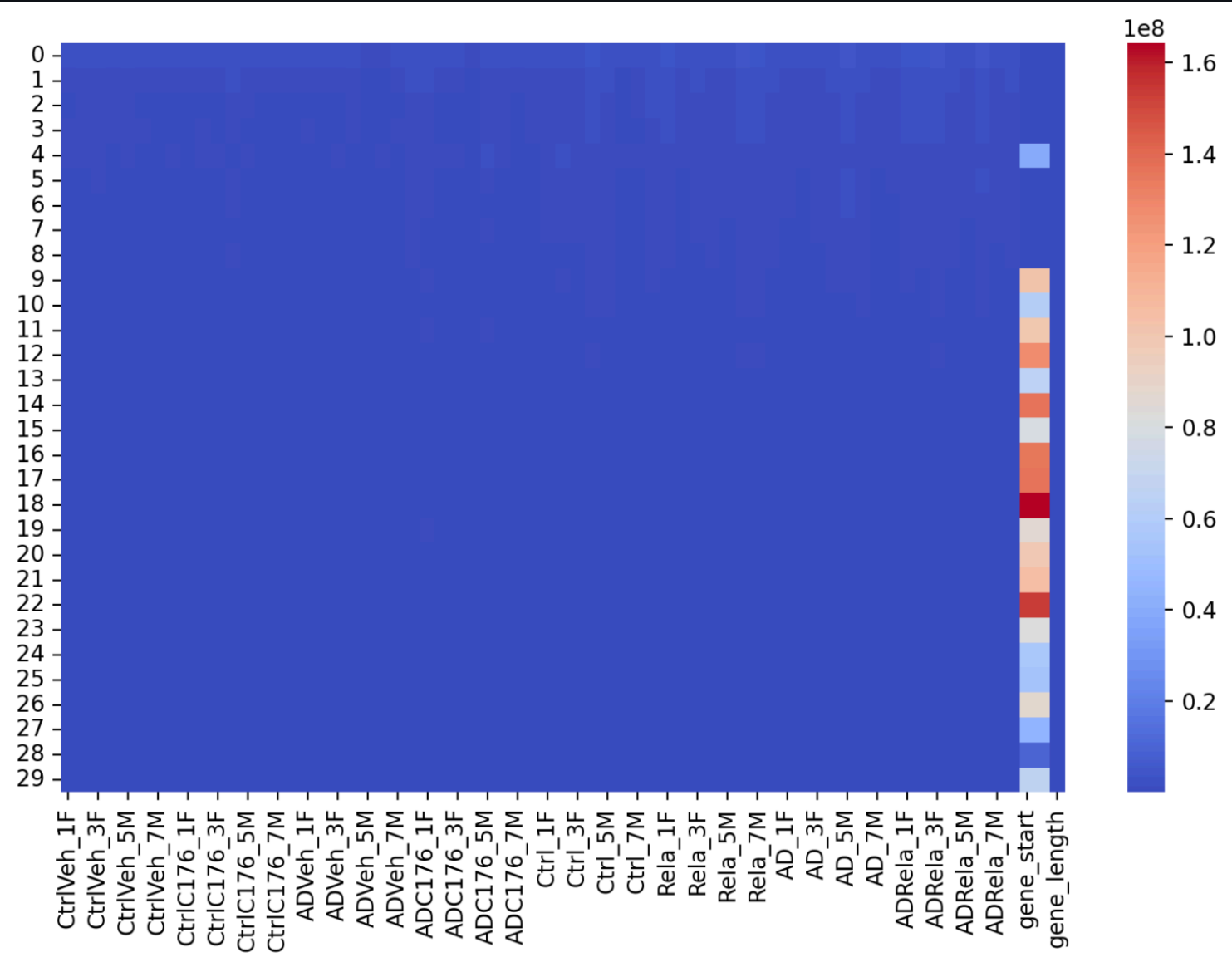
K-Means Cluster Visualization



INTERPRETATION:

- Each point represents a gene
- Genes with the same color belong to the same cluster
- Genes located close together show similar expression patterns

Gene Expression Heatmap

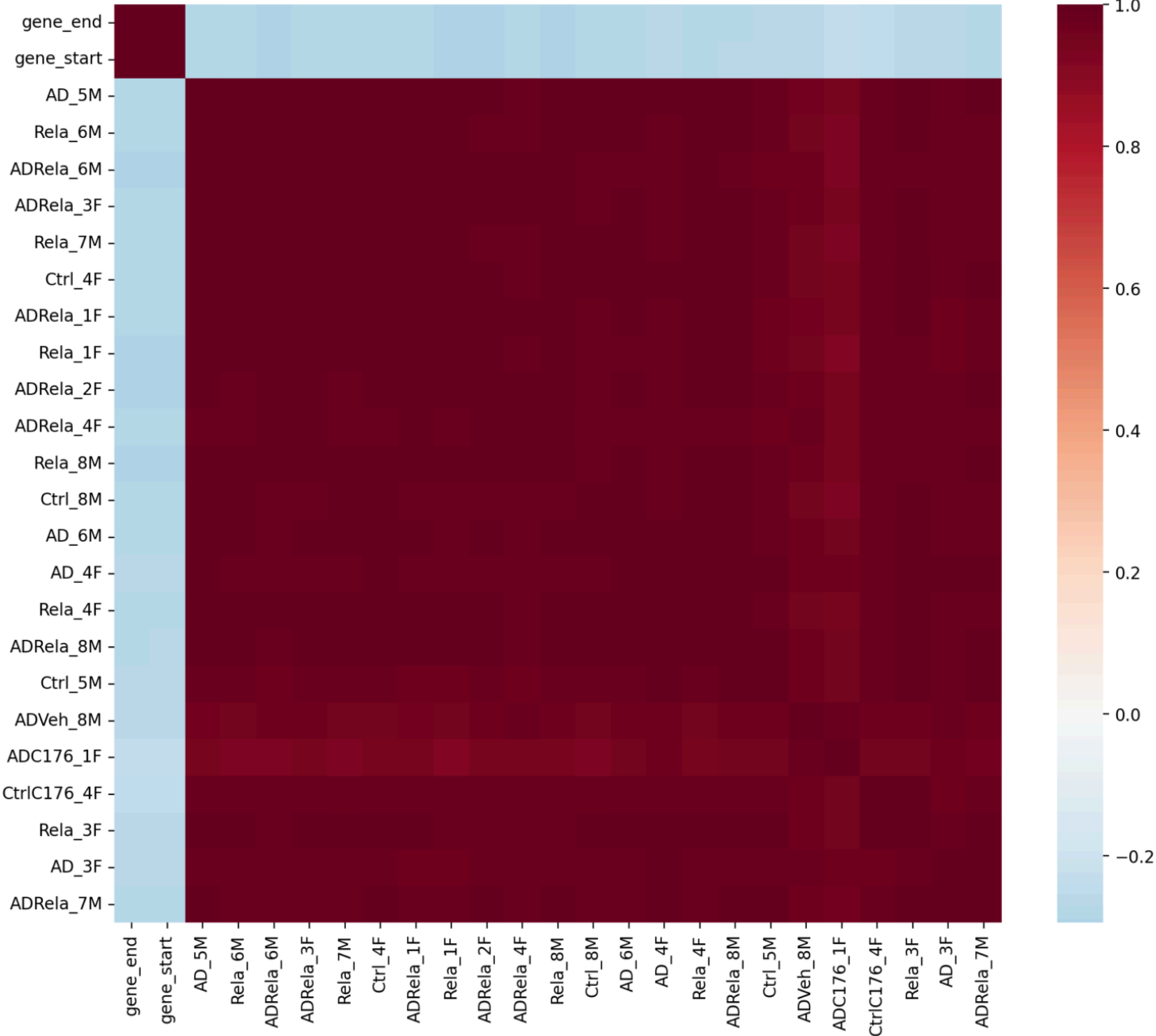


COLOR INTERPRETATION:

- Red indicates higher gene expression
- Blue indicates lower gene expression

- Similar color patterns suggest co-expressed genes

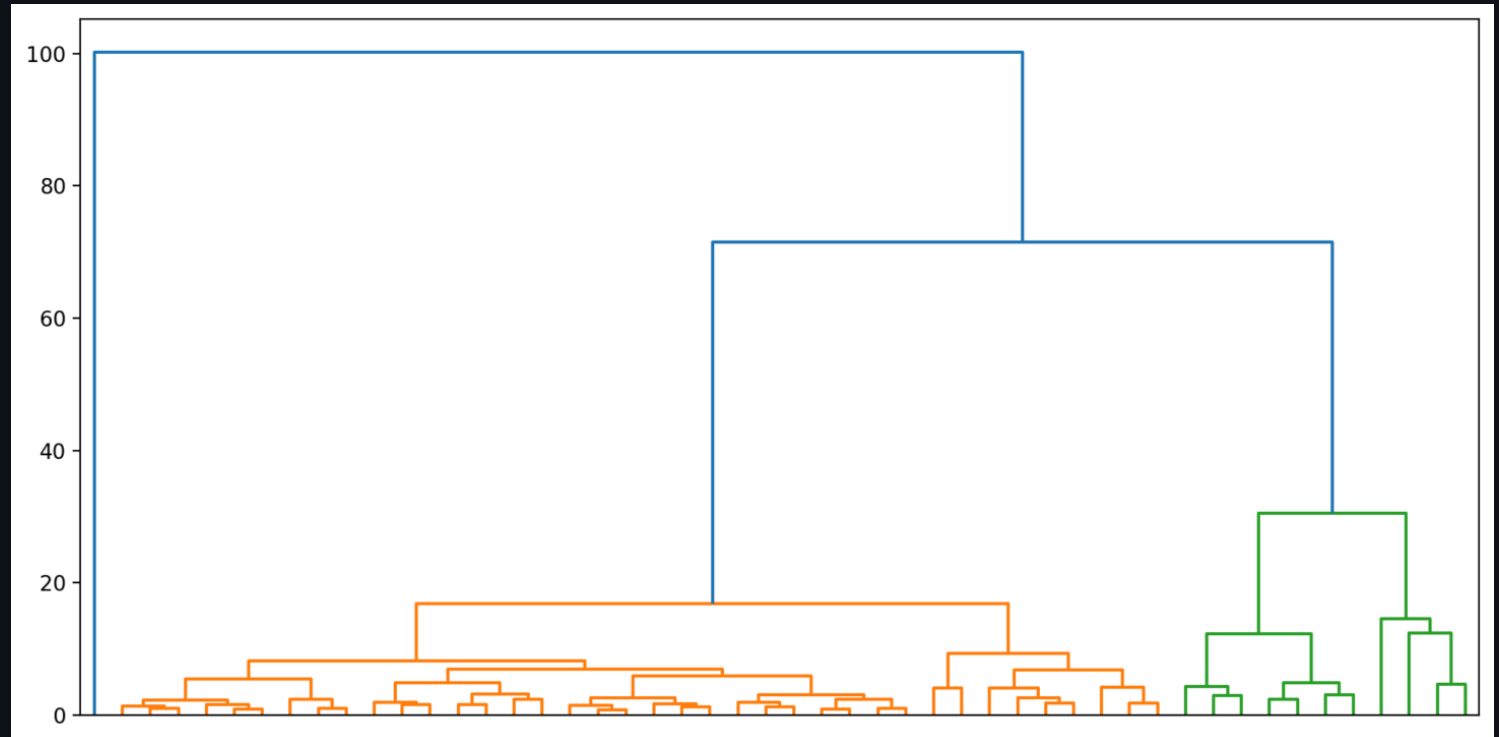
Gene Correlation Heatmap



COLOR INTERPRETATION:

- Red indicates positive correlation
- Blue indicates negative correlation
- Darker shades represent stronger relationships between genes

Hierarchical Clustering Dendrogram



DENDROGRAM INTERPRETATION:

- Shorter branches indicate stronger similarity between genes
- Longer branches indicate weaker similarity
- Closely connected branches represent gene clusters

Biological Interpretation

Genes within the same cluster showed similar expression patterns.

These co-expression patterns may indicate related biological pathways in Alzheimer's disease.

Generated Project Report

Project Report

PROJECT TITLE:
Gene Expression Clustering and Pattern Discovery

OBJECTIVE:
To analyze Alzheimer's gene expression data using clustering techniques and identify co-expression patterns.

- METHODOLOGY:
- Data preprocessing
 - Data normalization
 - PCA dimensionality reduction
 - K-Means clustering
 - Hierarchical clustering
 - Heatmap visualization

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